

Pretreatment Insulin Levels as a Prognostic Factor for Breast Cancer Progression

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Key Words. Breast cancer • Insulin • Insulin resistance • Prognostic value

ABSTRACT

Background. Based on the hypothesis that impaired glucose metabolism might be associated with survival outcomes independently of overt diabetes, we sought to investigate the prognostic value of routinely used glycemic parameters in a prospective study of breast cancer (BC) patients.

Patients and Methods. Fasting blood glucose, insulin and HbA_{1c} levels, and insulin resistance (assessed by the Homeostasis Model Assessment [HOMA] index) at diagnosis were evaluated in 286 nondiabetic BC patients (249 with primary cancer, 37 with metastatic) with respect to those parameters' possible associations with clinicopathological features and survival outcomes. As a control group, 143 healthy women matched in a 2:1 ratio for age, blood lipid levels, and body mass index were also investigated.

Results. Fasting blood glucose level (mean $\pm$ SD: 99 $\pm$ 26 vs. 85 $\pm$ 15 mg/dL), insulin level (median: 10.0 vs. 6.8 μ U/mL), and HOMA index (median: 2.2 vs. 1.4), but not HbA_{1c} level, were

significantly elevated in BC patients compared with control subjects. Receiver operating characteristics analysis showed comparable areas for blood glucose and insulin levels, and HOMA index (ranging from 0.668 to 0.671). Using a cutoff level of 13 μ U/mL, insulin had the best specificity (92%) and sensitivity (41%), was significantly associated with disease stage, and acted as a negative prognostic marker of progression-free survival (hazard ratio: 2.17; 95% confidence interval: 1.13–4.20) independently of menopausal status, disease stage, hormone receptor status, and human epidermal growth factor receptor 2 and Ki67 expression.

Conclusion. These results suggest that insulin determination might provide prognostic information in BC and support the hypothesis that lifestyle and/or pharmacological interventions targeting glucose metabolism could be considered to improve survival outcome of selected BC patients. *The Oncologist* 2016; 21:1041–1049

Implications for Practice: Pretreatment insulin levels may represent a biomarker of adverse prognosis in nondiabetic women with breast cancer, independently of other well-established prognostic factors (i.e., stage, hormone receptors, HER2/neu, and Ki67). This finding has important implications, because it provides the rationale for lifestyle or insulin-targeting pharmacologic interventions as a means of improving breast cancer outcomes not only in early stages, but also in advanced-stage breast cancer patients with aggressive tumor phenotypes (HER2-negative hormone-resistant, or triple-negative breast cancer), in which treatments are still challenging. The possibility of using insulin as a biomarker to guide insulin-targeted interventions also should be taken into account.

INTRODUCTION

General consensus exists on the possibility that type 2 diabetes (T2D) may contribute to breast cancer (BC) risk [1–6] and prognosis [7–9], but the underlying mechanisms are not completely understood. Hyperinsulinemia, insulin resistance (IR) [7, 8, 10], and enhanced levels of insulin-like growth factor-1 (IGF-1) [11] have all been proposed as the underlying

causative mechanisms, mainly on the basis of in vitro and animal studies supporting a role for insulin, insulin receptor, and IGFs in BC initiation [12–14] and progression [15]. These experimental findings are strengthened by indirect observations that metformin, a biguanide oral hypoglycemic agent capable of lowering insulin levels and improving IR, also might

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be effective in reducing BC incidence [8] and may beneficially act on proliferation and apoptosis markers in early BC stages [16–18]. Accordingly, metformin is being investigated in an adjuvant trial investigating invasive cancer-free survival in nondiabetic women with early breast cancer (National Cancer Institute of Canada Clinical Trials Group MA.32); the primary outcome measure is expected to be released late in 2017 [19].

Nevertheless, information on the prognostic value of insulin, or other glucose indexes, on survival outcomes is conflicting and there are no data, yet, supporting the role of these biomarkers in guiding insulin-targeted lifestyle or pharmacologic interventions as a way of improving BC outcomes. Most of the studies, in fact, have included only T2D patients [20–23] or patients with clinical features belonging to metabolic syndrome [24], including obesity [25–27] or dyslipidemia [28, 29], all leading to IR. Accordingly, the need for “improved study designs to address outstanding concerns surrounding the diabetes-cancer relationship” has been recently highlighted in a joint Consensus Statement of the American Association of Clinical Endocrinologists and the American College of Endocrinology [30].

Based on currently understood mechanisms for BC development in T2D, and aware of the inconsistent results reported in previous studies because of different study cohorts (e.g., choice of control subjects or mixed groups of patients with or without T2D), we decided to explore the potential prognostic value of routinely used glycemic parameters in women with BC in the absence of clinically diagnosed T2D. For this purpose, we first designed a case-control study aimed at evaluating the behavior of glucose metabolism indexes in nondiabetic women with BC compared with a healthy control population matched for age, body mass index (BMI), and blood lipid profile to minimize possible metabolic-related confounders. Based on the hypothesis that an association between pretreatment fasting blood glycemic indexes (blood glucose, insulin, and HbA_{1c} levels, and Homeostasis Model Assessment [HOMA] index) and BC-related survival outcomes might exist independently of the presence of an overt T2D, the prognostic value of routinely used glycemic parameters toward progression-free survival was then prospectively evaluated.

PATIENTS AND METHODS

Since January 2007, the Policlinico Tor Vergata Biospecimen Cancer Repository (PTV Bio.Ca.Re.) and the Interinstitutional Multidisciplinary Biobank of the IRCCS San Raffaele Pisana (SR-BioBIM) in Rome, Italy, have been actively involved in the recruitment of ambulatory patients with primary or metastatic cancer who are prospectively followed under the appropriate institutional ethics approval as part of a clinical database and biobank project. Among these, a cohort of 286 consecutive BC patients was selected for the study. Inclusion criteria for the BC patients from whom serum samples were stored in our biobanks were age older than 18 years, an Eastern Cooperative Oncology Group performance status of ≤ 2 , and adequate hematological, hepatic, and renal functions. T2D (HbA_{1c} level $\geq 6.5\%$, or fasting blood glucose level ≥ 126 mg/dL, or treatment with a hypoglycemic agent) or impaired glucose tolerance (IGT) (fasting blood glucose level: 100–125 mg/dL),

history of alcohol or drug abuse, and concurrent infectious or inflammatory diseases were all considered as exclusion criteria for the current analysis.

BC was staged according to the TNM classification. Breast surgery was performed in 249 women with primary BC (24% underwent mastectomy, 76% underwent lumpectomy). The remaining 37 patients had relapsing/metastatic disease and entered the study before the start of chemotherapy. Among the nonmetastatic population, 48 (19%) and 201 (81%) of the 249 women received neo-adjuvant and adjuvant therapies, respectively. Adjuvant chemotherapies, including those with anthracycline and those without, were instituted in 105 (52%) and 96 (48%) of the 201 patients with and without lymph node involvement, respectively. Among women with node-negative disease, 46 of 142 (32%) underwent adjuvant endocrine therapy only (tamoxifen or aromatase inhibitor). Patients with human epidermal growth factor receptor 2 (HER2/neu) positivity were all treated with trastuzumab-containing regimens. First-line chemotherapy was instituted in all patients with metastatic disease. Details on anticancer drugs used are reported in supplemental online Table 1. All patients were prospectively followed for a median of 3.92 years.

As a control group, 143 unrelated women from the same geographical area as the patients, paired for age, BMI, and blood lipid parameters, were recruited in a 2:1 ratio from apparently healthy individuals enrolled in the SR-BioBIM. Clinical and laboratory characteristics of patients and control subjects are summarized in Table 1.

The study was performed in accordance with the principles embodied in the Declaration of Helsinki. All patients gave written informed consent, previously approved by our institutional ethics committees.

Blood Sampling and Assessment of Glycemic Indexes

Fasting serum samples were obtained from each recruited subject, aliquoted, and stored at -80°C in the facilities of the PTV Bio.Ca.Re. or the SR-BioBIM. Samples from BC patients were obtained at baseline before any treatment.

Routine chemistry studies, including fasting blood glucose (hexokinase/glucose-6-phosphate dehydrogenase-based methodology; Abbott Laboratories, Abbott Park, IL, <http://www.abbott.com>), were performed with an ARCHITECT c8000 System (Abbott Laboratories) on fresh samples within 1 hour from blood withdrawal. Fasting insulin levels were analyzed on serum samples using a fully automated Lumipulse G 600 II chemiluminescent enzyme immunoassay analyzer (Fujirebio, Tokyo, Japan, <http://www.fujirebio.co.jp/english/index.html>) according to the manufacturer's instructions.

The HOMA index (a marker of insulin resistance) was retrospectively calculated for each participating subject from fasting blood glucose and insulin according to the following formula: $\text{glucose (mg/dL)} \times \text{insulin } (\mu\text{IU/mL}) / 405$ [31]. HbA_{1c} levels were measured with the Tosoh G7 Automated HPLC Analyzer – HbA_{1c} Variant Analysis Mode (Tosoh Bioscience, Rivoli, TO, Italy, <http://www.tosoh.com>), certified by the National Glycohemoglobin Standardization Program and traceable to the Diabetes Control and Complications Trial. All measurements were ascertained while blinded to the sample origin and to study endpoint.

Table 1. Clinical characteristics of breast cancer patients: comparison with control subjects

Characteristic	BC group (n = 286)	Control group (n = 143)
Age, mean $\pm$ SD (range), years	56 $\pm$ 13 (29–81)	55 $\pm$ 13 (28–80)
Menopausal status, n (%)		
Pre	130 (45)	67 (47)
Post	156 (55)	76 (53)
BMI, mean $\pm$ SD, kg/m ²	25.4 $\pm$ 4.8	25.3 $\pm$ 3.6
Blood lipid levels, mean $\pm$ SD, mg/dL		
Total cholesterol	213 $\pm$ 42	212 $\pm$ 40
HDL cholesterol	57 $\pm$ 14	56 $\pm$ 14
LDL cholesterol	132 $\pm$ 36	134 $\pm$ 34
Histological diagnosis, n (%)		
Ductal	251 (88)	—
Lobular	27 (9)	—
Other ^a	8 (3)	—
BC molecular type, n (%)		
Basal-like	40 (14)	—
Luminal-like	218 (76)	—
HER2 positive	28 (10)	—
Grading, n (%)		
1	17 (6)	—
2	92 (32)	—
3	177 (62)	—
Stage, n (%)		
I	69 (24)	—
II	112 (39)	—
III	64 (23)	—
IV ^b	4 (1)	—
Metastatic ^c	37 (13)	—
Hormone receptor status, n (%)		
ER + /PR +	206 (72)	—
ER + /PR –	27 (10)	—
ER – /PR +	6 (2)	—
ER – /PR –	47 (16)	—
HER2/neu +	28 (10)	—
Ki67 proliferation index $\geq$ 20%	204 (71)	—
Follow-up, median (range), years	3.92 (0.11–7.95)	—

^aIncluding mucinous (n = 5), and papillary, tubular, and comedocarcinoma (one each).

^bSingle metastatic site surgically resected.

^cDiagnosed in previously resected BC patients.

Abbreviations: —, not applicable; BC, breast cancer; BMI, body mass index; ER, estrogen receptor; HDL, high-density lipoprotein; HER2, human epidermal growth factor receptor 2; LDL, low-density lipoprotein; PR, progesterone receptor.

Assessment of Prognostic Indexes

Immunohistochemical analyses were performed on formalin-fixed, paraffin-embedded tumor sections for hormone (estrogen and progesterone) receptors [32], HER2/neu expression, and proliferation index (Ki67). HER2/neu positivity was defined according to the American Society of Clinical Oncology–College of American Pathologists guidelines as 3+ or 2+ immunohistochemical staining with evidence of gene amplification on fluorescence in situ hybridization [33]. Immunohistochemical detection of Ki67 was performed on the Ventana BenchMark XT automated staining platform (Roche Diagnostics, Mannheim, Germany, <http://www.roche.com>) according to manufacturer's

instructions. Ki67 proliferative index in surgical specimens was assigned by the pathologist on the basis of the percentage positive on at least 500 neoplastic cells counted in the peripheral area of the nodule. A cutoff value of $\geq$ 20% was used in all association analyses, according to the recommendations of the St. Gallen International Expert Consensus on the primary therapy of early BC 2013 [34].

Statistical Analysis

The sample size of the study was based on the agreement to inclusion criteria and willingness to provide informed consent, rather than on sample size calculations. However, estimation

Table 2. Glycemic parameters in breast cancer patients and control subjects

Parameter	BC group (n = 286)	Control group (n = 143)	p value
Fasting blood glucose, mean $\pm$ SD, mg/dL	99 $\pm$ 26	85 $\pm$ 15	<.0001
Fasting insulin, median (IQR), μ U/mL	10 (5.7–18.7)	6.8 (4.9–9.6)	<.0001
HbA _{1c} , mean $\pm$ SD, %	5.47 $\pm$ 0.54	5.34 $\pm$ 0.43	.061
HOMA index, median (IQR)	2.2 (1.2–4.3)	1.4 (1.0–2.0)	<.0001

Abbreviations: BC, breast cancer; HOMA, Homeostasis Model Assessment; IQR, interquartile range.

Table 3. Receiver operating characteristics and Bayesian analysis of glycemic parameters

Characteristics	Fasting blood glucose	Fasting insulin	HOMA index
AUC (SE)	0.669 (0.03)	0.668 (0.03)	0.671 (0.03)
95% CI	0.619–0.717	0.618–0.715	0.617–0.722
Cutoff level	97 mg/dL	13 μ U/mL	3
Sensitivity, %	39	41	40
Specificity, %	88	92	92
Accuracy, %	52	53	54
OR (95% CI)	3.4 (1.9–6.1)	6.3 (3.0–13.3)	5.7 (2.9–11.4)
p value	.0001	.0001	.0001

Abbreviations: AUC, area under the curve; CI, confidence interval; HOMA, Homeostasis Model Assessment; OR, odds ratio.

was later performed and showed that, given the observed means of patients and control groups for insulin values and using a type I error probability of .05, the recruited population yielded a statistical power greater than 95%.

Data are presented as percentages, mean $\pm$ SD, or median and interquartile range. Student's unpaired *t* test and analysis of variance (ANOVA) were used for normally distributed variables. Appropriate nonparametric tests (Mann-Whitney U and Kruskal-Wallis ANOVA and median tests) were used for all the other variables. The cutoff values were generated from continuous data by ROC curve analyses performed with MedCalc Statistical Software version 13.1.2 (MedCalc Software, Ostend, Belgium, <http://www.medcalc.org>).

Progression-free survival (PFS), representing the study endpoint, was calculated from the date of enrollment until relapse or progression of disease. If a patient had not progressed or died, PFS was censored at the time of the last follow-up. PFS curves were calculated by the Kaplan-Meier method and the significance level was assessed according to the log-rank test using a computer software package (Statistica 8.0; StatSoft, Tulsa, OK, <http://www.statsoft.com>). Cox proportional hazards analysis was performed by a free Internet-based application (<http://statpages.org>) to evaluate the association between clinicopathological variables and PFS. For administrative censoring, follow-up was ended on March 31, 2015. All tests were two-tailed and only *p* values less than .05 were regarded as statistically significant.

RESULTS

Of 329 prospectively recruited BC patients, 34 (10%) had an established diagnosis of T2D and were excluded from the analysis. Fasting blood glycemic indexes (blood glucose, insulin, and HbA_{1c} levels, and HOMA index) were retrospectively reviewed, demonstrating the presence of an impaired glucose tolerance in 9 patients (3%); these 9 were also excluded, leaving a total of 286 BC patients eligible for analysis. A diagram of the

patients' recruitment is depicted in supplemental online Figure 1. No patient was lost to follow-up.

Fasting glucose (*p* < .0001) and insulin (*p* < .0001), but not HbA_{1c}, levels were higher in nondiabetic patients with breast cancer compared with nondiabetic patients without breast cancer (Table 2). Of interest, median pretreatment insulin levels increased with BC stage (stage I: 8.1 μ U/mL; stage II: 8.6 μ U/mL; stage III: 12.2 μ U/mL), and were highest in patients with metastatic disease (median insulin level: 16.7 μ U/mL; Kruskal-Wallis test: *H* = 20.4, *p* = .0004), whereas fasting glucose levels did not differ across early stages, but did differ in comparison with metastatic disease (Kruskal-Wallis test: *H* = 9.7, *p* = .045) (supplemental online Fig. 2). Accordingly, HOMA index was increased in patients with BC in association with the stage of disease. Given the lack of association with BC, HbA_{1c} levels were excluded from subsequent analyses.

As summarized in Table 3, the areas under the curve for fasting blood glucose and insulin levels and HOMA index were: 0.669, 0.668, and 0.671, respectively. Bayesian analysis further demonstrated that insulin level (cutoff: 13 μ U/mL) and HOMA index (cutoff: 3) had the highest sensitivities and specificities: They were associated with a sixfold higher risk for BC.

The relationship between glycemic indexes and clinicopathological features of BC was further analyzed after variable categorization. Cross-tabulation analyses are reported in supplemental online Table 2, confirming the association between elevated pretreatment glucose level (*p* = .001), insulin level (*p* < .0001), or HOMA index (*p* < .0001) and stage of disease. No associations were observed between glycemic indexes and any of the BC prognostic factors analyzed in this study.

Overall, the median follow-up was 3.92 years. Among patients with primary BC, 219 of the 249 (88%) remained clinically free of disease, and 30 of the 249 (12%) had recurrence or new primary breast cancer (one second primary BC after 1 year). Among patients with relapsing metastatic BC, 1 had stable disease, 9 of 37 (24%) had a complete or partial response during

Table 4. Cox proportional hazards survival regression analysis of the predictive value of clinicopathological variables and glycemic indexes on progression-free survival of breast cancer patients

Variable	N	Progression, n (%)		HR (95% CI)	p value
		Yes	No		
Menopausal status					
Pre	130	35 (27)	95 (73)		
Post	156	22 (14)	134 (86)	0.60 (0.34–1.04)	.067
Histological diagnosis					
Ductal	251	50 (20)	201 (80)		
Lobular	27	7 (26)	20 (74)		
Others	8	0 (0)	8 (100)	0.88 (0.44–1.75)	.721
Stage of disease					
I	69	5 (7)	64 (93)		
II	112	15 (13)	97 (87)		
III	64	8 (13)	56 (88)		
IV	4	2 (50)	2 (50)		
Metastatic	37	27 (73)	10 (27)	1.88 (1.46–2.42)	.0001 ^a
Hormone receptor status					
ER+/PR+	206	42 (20)	164 (80)		
ER+/PR–	27	1 (4)	26 (96)		
ER–/PR+	6	3 (50)	3 (50)		
ER–/PR–	47	11 (23)	36 (77)	1.11 (0.86–1.42)	.424
HER2/neu status					
Negative	258	49 (20)	209 (81)		
Positive	28	8 (29)	20 (71)	1.45 (0.65–3.25)	.364
Ki67 proliferation index					
Low	82	11 (13)	71 (87)		
High	204	46 (23)	158 (77)	1.17 (0.57–2.41)	.665
Body mass index, kg/m ²					
≤25	157	26 (17)	131 (83)		
25–30	89	22 (25)	67 (75)		
>30	40	9 (23)	31 (77)	0.98 (0.68–1.41)	.898
Type of treatment					
Neoadjuvant	48	11 (23)	37 (77)		
Hormone CT	48	2 (4)	46 (96)		
Adjuvant CT	154	17 (11)	137 (89)		
Metastatic	37	27 (73)	10 (27)	1.21 (0.89–1.64)	.236
Fasting glucose level					
Low	184	32 (17)	152 (83)		
High	102	25 (24)	77 (76)	0.82 (0.44–1.51)	.515
Fasting insulin level					
Low	191	23 (12)	168 (88)		
High	95	34 (36)	61 (64)	2.17 (1.13–4.20)	.021 ^a

^aStatistically significant ($p < .05$).

Abbreviations: CI, confidence interval; CT, chemotherapy; ER, estrogen receptor; HER2, human epidermal growth factor receptor 2; HR, hazard ratio; PR, progesterone receptor.

chemotherapy, and 27 (73%) had BC progression. Elevated insulin levels and impaired HOMA index, but not fasting glycemia, were approximately two times higher in patients who had BC progression compared with patients in whom the disease did not progress (supplemental online Table 2).

Univariate Cox proportional hazards survival analyses showed that elevated pretreatment insulin levels (HR: 3.41;

95% CI: 2.01–5.79), but not blood glucose level (HR: 1.65; 95% CI: 0.97–2.78), had a negative prognostic value in terms of PFS. These results were largely unmodified after adjustment for other variables known to be associated with PFS, including disease stage, hormone receptor and HER2/neu status, and Ki67 proliferation index (HR for insulin levels $>13 \mu\text{IU/mL}$: 2.17; 95% CI: 1.13–4.20; overall model fit: $p < .0001$) (Table 4).

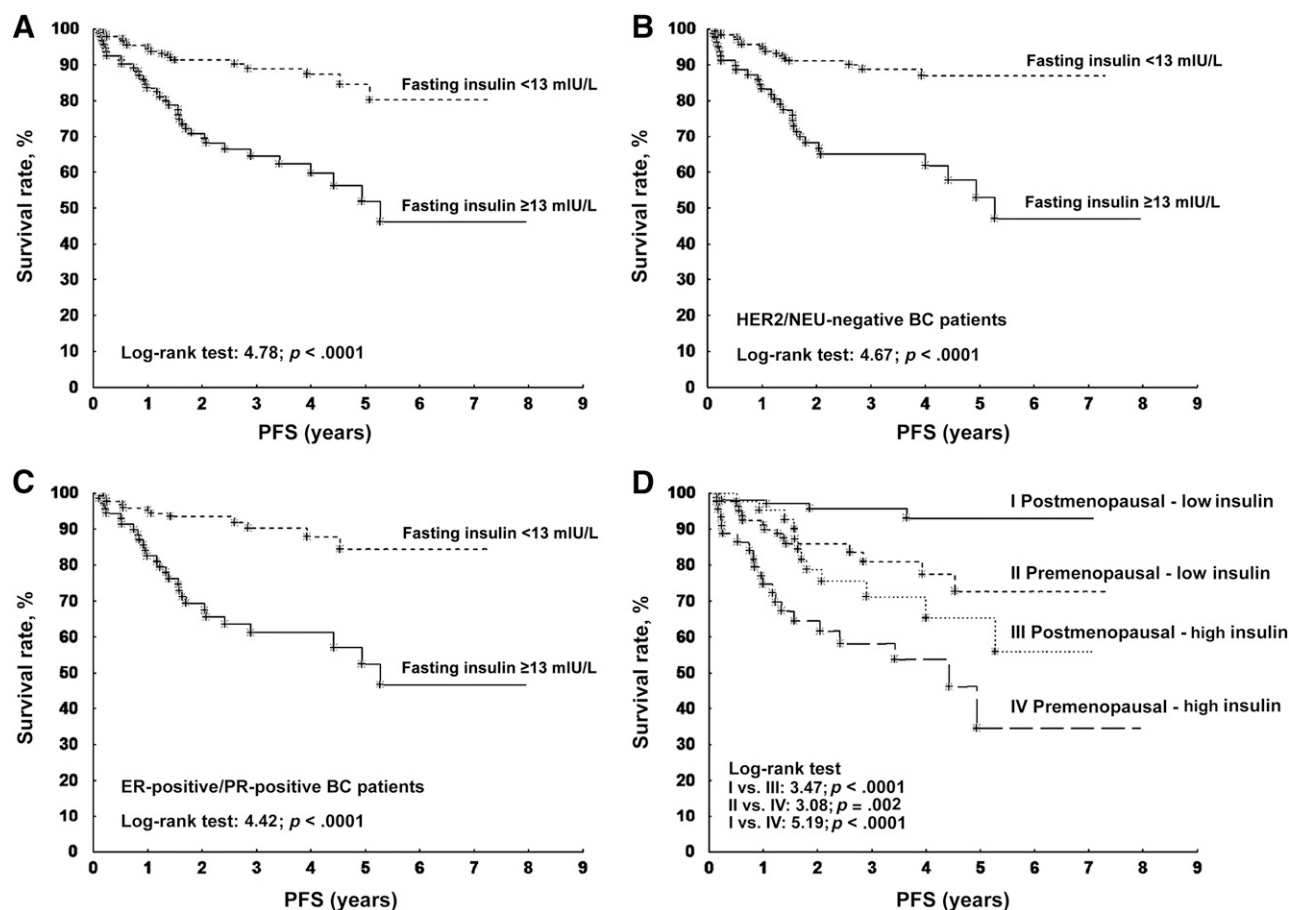


Figure 1. Kaplan-Meier curves of PFS of patients with breast cancer. Comparison between patients with low (<13 μ U/mL) and high ($\geq$ 13 μ U/mL) fasting insulin levels. (A): Overall population. (B): Patients with negative HER2/neu expression. (C): Patients with positive hormone receptors. (D): Association with menopausal status.

Abbreviations: BC, breast cancer; ER, estrogen receptor; PFS, progression-free survival; PR, progesterone receptor.

Figure 1A demonstrates the Kaplan-Meier PFS curves for BC patients stratified on the basis of pretreatment insulin levels. As shown, patients with insulin levels $>13 \mu$ U/mL had a worse 5-year survival rate compared with patients with insulin levels below this cutoff (55% vs. 85%; log-rank: 4.8; $p < .0001$). Similar results were confirmed in a subset of patients with primary nonmetastatic BC when analyzed for progression-free survival (log-rank: 3.6; $p = .0003$) (supplemental online Fig. 3).

Figure 1B–1D shows the Kaplan-Meier PFS curves of patients with negative HER2/neu status (Fig. 1B), positive hormone receptors (Fig. 1C), or menopausal status (Fig. 1D) categorized on the basis of insulin levels. Of interest, postmenopausal women with pretreatment insulin levels $<13 \mu$ U/mL had the most favorable survival outcome (5-year PFS rate: 92%), whereas premenopausal women with insulin levels above this cutoff had the worst survival (5-year PFS rate: 36%; log-rank = 5.87; $p < .0001$).

The negative prognostic value of insulin was further confirmed in subgroup analyses of patients categorized according to the main prognostic clinicopathological variables. As summarized in Table 5, patients with pretreatment insulin levels $>13 \mu$ U/mL had worse 5-year survival rates irrespective of hormone receptor status or Ki67 proliferation index. On the other hand, prognostic value was lost in patients with metastatic disease or HER2/neu positivity.

DISCUSSION

Despite the availability of a large amount of data on the presence of a causal link between IGT or T2D and BC risk [35–37], the role of sustained hyperglycemia, hyperinsulinemia, or IR in BC prognosis is far less investigated [38], especially in nondiabetic BC patients. Indeed, only a few studies—mostly from a single research group—have been performed in cohorts of BC patients with metabolic characteristics comparable to those of the general population, while the majority of the articles available in the international literature have focused on the association between BC and glucose metabolism in patients with clinical features belonging to metabolic syndrome [24, 39], obesity [25–27, 40] or dyslipidemia [28, 29], all of which recognize IR as a common factor.

Here, we report the results obtained in a cohort of nondiabetic women with BC, demonstrating that BC patients had increased, albeit not pathological, pretreatment fasting blood glucose and insulin levels compared with control subjects matched for age, BMI, and lipid profile, with the higher pretreatment levels resulting in higher HOMA indexes. A condition of IR (HOMA index >2.5) was found in 36% of normal weight and 50%–61% of overweight/obese BC patients, respectively ($p = .039$). Nonetheless, multivariate regression

Table 5. Rates of 5-year progression-free survival of breast cancer patients according to main prognostic clinicopathological variables

	Fasting insulin, %		Log-rank test	p value
Variable	High	Low		
Stage of disease				
Localized	67	84	3.91	<.0001
Metastatic	18	18	0.03	.977
Hormone receptor status				
ER+/PR+	52	81	4.42	<.001
ER-/PR-	31	82	1.53	.126
HER2/neu status				
Negative	53	84	4.67	<.0001
Positive	42	43	0.84	.403
Ki67 proliferation index				
Low	70	94	2.35	.019
High	42	79	4.48	<.0001

Abbreviations: ER, estrogen receptor; HER2, human epidermal growth factor receptor 2; PR, progesterone receptor.

analyses specifically designed to analyze the major determinants of glycemic indexes, including menopausal status, disease stage, BMI, and blood lipids, showed that BMI was not an independent predictor of either insulin level or HOMA index in our patient cohort (supplemental online Table 3), suggesting that BMI would not represent a confounding factor in association analyses between insulin level and BC clinical outcome. Finally, HbA_{1c} concentrations did not differ between patients and control subjects, and only 4 patients (1.4%) had isolated HbA_{1c} levels between 6.5% and 7.0%, which were not reproduced at second testing, thus confirming, in agreement with the most recent guidelines for T2D diagnostic criteria [41], that the differences observed were not due to unrecognized diabetes in our study population.

The major findings of our study were a significant association between pretreatment insulin levels and advanced disease stage, translating to a positive association of HOMA index, and the demonstration that pretreatment insulin levels (HR: 2.2) and HOMA index (HR: 1.91; 95% CI: 1.06–3.42) (data not shown) might act as negative prognostic factors for PFS, independently of other well-established prognostic factors (i.e., stage, hormone receptors, HER2/neu and Ki67 expression). Indeed, patients with insulin levels above a cutoff of 13 μ U/mL had a twofold increased risk for disease progression (5-year survival rate: 55%) compared with patients with insulin levels below this cutoff (5-year survival rate: 85%), corroborating the experimental findings of an insulin role in tumor growth and metastasis either directly, by stimulation of cell proliferation and/or inhibition of apoptosis [2, 13], or indirectly through the elevation of endogenous sex steroid hormones caused by insulin-mediated reduction of hepatic sex hormone-binding globulin [6].

The hypothesis that insulin might represent a mediator of adverse prognosis in nondiabetic women with BC had been advocated by Goodwin et al. [42, 43], who suggested that patients with locoregional BC and fasting insulin levels in the

upper quartile ($>7.8 \mu$ U/mL) had a twofold increased risk for distant recurrence compared with those in the lower quartile ($<4.0 \mu$ U/mL). According to Goodwin et al. [42, 43], the high insulin levels found in BC reflected an underlying IR state because they were associated with other components of the IR syndrome, namely BMI, waist circumference, or waist-to-hip ratio. In favor of their hypothesis, these authors also reported significant correlations between fasting insulin and triglyceride levels, or high density lipoprotein cholesterol levels, both additional components of the IR syndrome.

Although our results are in agreement with the hypothesis of a negative prognostic value of insulin, they are partially divergent from those obtained in the study by Goodwin et al., because neither overweight/obesity (assessed by BMI) nor blood lipids were independent predictors of insulin levels at multivariate analysis, with the only exception being a borderline association with triglycerides (supplemental online Table 3). This might be partially explained by differences in the populations recruited, because the studies performed by Goodwin and coworkers included younger women (approximately 61% premenopausal) with early BC stages [42–44], whereas our population was older (55% postmenopausal) and encompassed all stages of disease, including metastatic disease. In fact, postmenopausal status (regression coefficient: 0.193; $p < .001$) and stage of disease (regression coefficient: 0.168; $p = .02$) were the sole independent predictors of fasting glucose level, which, in turn, was the strongest predictor of insulin level (supplemental online Table 3), a finding that is in agreement with a previous report demonstrating that menopause, when the hormonal and metabolic imbalances become stronger, but not age, is directly involved in augmented fasting blood glucose levels in non-diabetic women [45].

Despite the observed differences, the results of our study are in agreement with, and extend, those reported by Goodwin et al. [42, 43], suggesting that pretreatment insulin levels might have a prognostic role not only in early BC but also in advanced stages of disease, thus reinforcing the rationale for lifestyle or insulin-targeting pharmacologic interventions as a means of improving breast cancer outcomes. In this light, it is worth mentioning an *in vitro* study on BC cells showing how insulin priming potentially contributes to the estradiol-induced cancer growth by modulating estradiol-insulin signaling crosstalk [46].

A further implication for BC clinical management that might arise from the present study is the possibility of using pretreatment insulin level as a biomarker to guide insulin-targeted interventions in selected groups of BC patients. It is well known that hormone receptor status and HER2/neu expression are associated with patients' outcome. In particular, the lack of expression of estrogen and progesterone receptors has been associated with poorer prognosis [47], and HER2 overexpression has been associated with increased recurrence and mortality [48]. In this context, Goodwin et al. [43] reported that the prognostic association for disease-free survival of obesity-related variables (including insulin) did not differ in women with hormone receptor-positive versus -negative cancers, suggesting that insulin-targeting interventions should be evaluated in all patients independently of hormone receptor status [43]. These findings were in disagreement with those reported by other groups suggesting that obesity and BMI at diagnosis are associated with inferior outcomes in

hormone receptor-positive BC only [49, 50]. In agreement with the latter studies, the results of our study seem to suggest that the prognostic value of insulin is maximized in hormone receptor-positive BC. However, the small sample size of hormone receptor-negative patients might have caused an underestimate of the prognostic significance of pretreatment insulin levels in this subset of patients.

On the other hand, and to best of our knowledge, this is the first report suggesting that pretreatment insulin levels might have a prognostic value in HER2-negative patients. The use of a 13 μ U/mL cutoff for insulin was capable of discriminating a subset of women with a more favorable outcome, with a 5-year progression-free survival rate of 87% (median time to progression: 34 months) in patients with low pretreatment insulin levels compared with 52% of BC patients with elevated insulinemia (median time to progression: 14 months; $p \leq .0001$). These results are barely comparable with the currently available clinical and experimental evidence, because there are no data on the prognostic value of insulin level or IR in HER2-negative BC. The few data available point to a higher prevalence of metabolic syndrome [51] or obesity [52] in triple-negative BC (TNBC) patients as opposed to non-triple-negative patients; it has been advocated that obesity-related defective estrogen surveillance might allow steroid receptor-negative BC to more easily escape detection than steroid receptor-positive BC [52]. Meanwhile, it has been shown that metformin is able to inhibit proliferation and colony formation of TNBC cells in vitro [53], a finding that was confirmed in mice models in which metformin treatment resulted in decreased growth of TNBC xenografts and decreased tumor formation if administered before tumor cell injection [54]. In agreement with these findings, elevated pretreatment insulin levels in TNBC patients enrolled in our study were associated with a worse PFS (5-year survival rate: 33%; median time to progression: 13 months) compared with TNBC patients with low insulin levels (5-year survival rate: 80%; median time to progression: 34 months; $p = .002$), but the rate of TNBC was too small and the statistical power too low to draw any conclusion (HR: 2.27; 95% CI: 0.63–8.20). Nonetheless, the findings reported here allow us to speculate that lifestyle interventions and insulin-targeting drugs, namely metformin, may prove beneficial not only for the treatment of early BC stages, as recently suggested on the basis of the positive metabolic effects of metformin in this particular cluster of patients [19], but also in subsets of patients characterized by more aggressive tumor phenotypes, such as HER2-negative hormone-resistant tumors, or TNBC, in which treatments are still challenging.

There are, of course, some limitations to our study that need to be acknowledged. The most obvious resides in the

relatively small sample size that might have weakened the statistical power, leading to borderline significance. The strength of our analysis is represented by the use of samples collected and processed using standard operating procedures in the context of two large biobanks. In addition, tests were run by a single laboratory under ongoing quality control protocols, which minimized the difference in sample analyses.

CONCLUSION

Pretreatment insulin levels might have a negative prognostic value in BC patients. These results, however, should be regarded with caution and detailed experimental evaluation is needed before the ultimate prognostic significance of insulin in breast cancer can be established. Additional studies are required to prospectively evaluate the clinical value of pretreatment insulin levels in BC. Nevertheless, we believe that its determination could provide important information in risk stratification, and we encourage future investigations addressing the role of insulin in the management of BC patients and in providing the rationale for new therapeutic strategies.

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DISCLOSURES

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REFERENCES

- Larsson SC, Mantzoros CS, Wolk A. Diabetes mellitus and risk of breast cancer: A meta-analysis. *Int J Cancer* 2007;121:856–862.
- Giovannucci E, Harlan DM, Archer MC et al. Diabetes and cancer: A consensus report. *CA Cancer J Clin* 2010;60:207–221.
- Chodick G, Zucker I. Diabetes, gestational diabetes and the risk of cancer in women: Epidemiologic evidence and possible biologic mechanisms. *Womens Health (Lond Engl)* 2011; 7:227–237.
- Vona-Davis L, Rose DP. Type 2 diabetes and obesity metabolic interactions: Common factors for breast cancer risk and novel approaches to prevention and therapy. *Curr Diabetes Rev* 2012;8:116–130.
- Miniccozi P, Berrino F, Sebastiani F et al. High fasting blood glucose and obesity significantly and independently increase risk of breast cancer death in hormone receptor-positive disease. *Eur J Cancer* 2013;49:3881–3888.
- Joung KH, Jeong JW, Ku BJ. The association between type 2 diabetes mellitus and women cancer: the epidemiological evidences and putative mechanisms. *Biomed Res Int* 2015;2015:920618.
- Pisani P. Hyper-insulinaemia and cancer, meta-analyses of epidemiological studies. *Arch Physiol Biochem* 2008;114:63–70.
- Decensi A, Puntoni M, Goodwin P et al. Metformin and cancer risk in diabetic patients: A systematic review and meta-analysis. *Cancer Prev Res (Phila)* 2010;3:1451–1461.
- Formica V, Tesaro M, Cardillo C et al. Insulinemia and the risk of breast cancer and its

relapse. *Diabetes Obes Metab* 2012;14:1073–1080.

10. Gunter MJ, Hoover DR, Yu H et al. Insulin, insulin-like growth factor-I, and risk of breast cancer in postmenopausal women. *J Natl Cancer Inst* 2009; 101:48–60.

11. Duggan C, Wang CY, Neuhaus ML et al. Associations of insulin-like growth factor and insulin-like growth factor binding protein-3 with mortality in women with breast cancer. *Int J Cancer* 2013;132:1191–1200.

12. LeRoith D. Can endogenous hyperinsulinaemia explain the increased risk of cancer development and mortality in type 2 diabetes: Evidence from mouse models. *Diabetes Metab Res Rev* 2010; 26:599–601.

13. Rose DP, Vona-Davis L. The cellular and molecular mechanisms by which insulin influences breast cancer risk and progression. *Endocr Relat Cancer* 2012;19:R225–R241.

14. Arcidiacono B, Iiritano S, Nocera A et al. Insulin resistance and cancer risk: An overview of the pathogenetic mechanisms. *Exp Diabetes Res* 2012; 2012:789174.

15. Gallagher EJ, Alikhani N, Tobin-Hess A et al. Insulin receptor phosphorylation by endogenous insulin or the insulin analog AspB10 promotes mammary tumor growth independent of the IGF-I receptor. *Diabetes* 2013;62:3553–3560.

16. Hadad S, Iwamoto T, Jordan L et al. Evidence for biological effects of metformin in operable breast cancer: A pre-operative, window-of-opportunity, randomized trial. *Breast Cancer Res Treat* 2011;128: 783–794.

17. Bonanni B, Puntoni M, Cazzaniga M et al. Dual effect of metformin on breast cancer proliferation in a randomized presurgical trial. *J Clin Oncol* 2012;30: 2593–2600.

18. Niraula S, Dowling RJ, Ennis M et al. Metformin in early breast cancer: A prospective window of opportunity neoadjuvant study. *Breast Cancer Res Treat* 2012;135:821–830.

19. Goodwin PJ, Parulekar WR, Gelmon KA et al. Effect of metformin vs placebo on end and metabolic factors in NCIC CTG MA.32. *J Natl Cancer Inst* 2015; 107:djv006.

20. Jiralerspong S, Kim ES, Dong W et al. Obesity, diabetes, and survival outcomes in a large cohort of early-stage breast cancer patients. *Ann Oncol* 2013; 24:2506–2514.

21. Hou G, Zhang S, Zhang X et al. Clinical pathological characteristics and prognostic analysis of 1,013 breast cancer patients with diabetes. *Breast Cancer Res Treat* 2013;137:807–816.

22. Wu AH, Kurian AW, Kwan ML et al. Diabetes and other comorbidities in breast cancer survival by race/ethnicity: The California Breast Cancer Survivorship Consortium (CBCSC). *Cancer Epidemiol Biomarkers Prev* 2015;24:361–368.

23. Xiao Y, Zhang S, Hou G et al. Clinical pathological characteristics and prognostic analysis of diabetic women with luminal subtype breast cancer. *Tumour Biol* 2014;35:2035–2045.

24. Calip GS, Malone KE, Gralow JR et al. Metabolic syndrome and outcomes following early-stage breast cancer. *Breast Cancer Res Treat* 2014;148:363–377.

25. Parekh N, Lin Y, Vadiveloo M et al. Metabolic dysregulation of the insulin-glucose axis and risk of obesity-related cancers in the Framingham heart study-offspring cohort (1971-2008). *Cancer Epidemiol Biomarkers Prev* 2013;22:1825–1836.

26. Chan DS, Vieira AR, Aune D et al. Body mass index and survival in women with breast cancer-systematic literature review and meta-analysis of 82 follow-up studies. *Ann Oncol* 2014;25:1901–1914.

27. Gunter MJ, Xie X, Xue X et al. Breast cancer risk in metabolically healthy but overweight postmenopausal women. *Cancer Res* 2015;75:270–274.

28. Bahl M, Ennis M, Tannock IF et al. Serum lipids and outcome of early-stage breast cancer: Results of a prospective cohort study. *Breast Cancer Res Treat* 2005;94:135–144.

29. Bell KE, Di Sebastiano KM, Vance V et al. A comprehensive metabolic evaluation reveals impaired glucose metabolism and dyslipidemia in breast cancer patients early in the disease trajectory. *Clin Nutr* 2014;33:550–557.

30. Handelsman Y, Leroith D, Bloomgarden ZT et al. Diabetes and cancer—an AACE/ACE consensus statement. *Endocr Pract* 2013;19:675–693.

31. Bonora E, Targher G, Alberiche M et al. Homeostasis model assessment closely mirrors the glucose clamp technique in the assessment of insulin sensitivity: Studies in subjects with various degrees of glucose tolerance and insulin sensitivity. *Diabetes Care* 2000;23:57–63.

32. Hammond ME, Hayes DF, Dowsett M et al. American Society of Clinical Oncology/College of American Pathologists guideline recommendations for immunohistochemical testing of estrogen and progesterone receptors in breast cancer. *J Clin Oncol* 2010;28:2784–2795.

33. Wolff AC, Hammond ME, Schwartz JN et al. American Society of Clinical Oncology/College of American Pathologists guideline recommendations for human epidermal growth factor receptor 2 testing in breast cancer. *Arch Pathol Lab Med* 2007; 131:18–43.

34. Goldhirsch A, Winer EP, Coates AS et al. Personalizing the treatment of women with early breast cancer: Highlights of the St Gallen International Expert Consensus on the Primary Therapy of Early Breast Cancer 2013. *Ann Oncol* 2013;24: 2206–2223.

35. Pollak M. The insulin and insulin-like growth factor receptor family in neoplasia: An update. *Nat Rev Cancer* 2012;12:159–169.

36. Orgel E, Mittelman SD. The links between insulin resistance, diabetes, and cancer. *Curr Diab Rep* 2013;13:213–222.

37. Noto H, Goto A, Tsujimoto T et al. Latest insights into the risk of cancer in diabetes. *J Diabetes Investig* 2013;4:225–232.

38. Peairs KS, Barone BB, Snyder CF et al. Diabetes mellitus and breast cancer outcomes: A systematic review and meta-analysis. *J Clin Oncol* 2011;29:40–46.

39. Emaus A, Veierød MB, Tretli S et al. Metabolic profile, physical activity, and mortality in breast cancer patients. *Breast Cancer Res Treat* 2010;121:651–660.

40. Duggan C, Irwin ML, Xiao L et al. Associations of insulin resistance and adiponectin with mortality in women with breast cancer. *J Clin Oncol* 2011;29:32–39.

41. American Diabetes Association. 2. Classification and diagnosis of diabetes. *Diabetes Care* 2015; 38(suppl 1):S8–S16.

42. Goodwin PJ, Ennis M, Pritchard KI et al. Fasting insulin and outcome in early-stage breast cancer: Results of a prospective cohort study. *J Clin Oncol* 2002;20:42–51.

43. Goodwin PJ, Ennis M, Pritchard KI et al. Insulin- and obesity-related variables in early-stage breast cancer: Correlations and time course of prognostic associations. *J Clin Oncol* 2012;30:164–171.

44. Goodwin PJ, Ennis M, Bahl M et al. High insulin levels in newly diagnosed breast cancer patients reflect underlying insulin resistance and are associated with components of the insulin resistance syndrome. *Breast Cancer Res Treat* 2009;114:517–525.

45. Otsuki M, Kasayama S, Morita S et al. Menopause, but not age, is an independent risk factor for fasting plasma glucose levels in nondiabetic women. *Menopause* 2007;14:404–407.

46. Wairagu PM, Phan ANH, Kim MK et al. Insulin priming effect on estradiol-induced breast cancer metabolism and growth. *Cancer Biol Ther* 2015;16: 484–492.

47. Hayes DF. Prognostic and predictive factors revisited. *Breast* 2005;14:493–499.

48. Kaplan HG, Malmgren JA. Impact of triple negative phenotype on breast cancer prognosis. *Breast J* 2008;14:456–463.

49. de Azambuja E, McCaskill-Stevens W, Francis P et al. The effect of body mass index on overall and disease-free survival in node-positive breast cancer patients treated with docetaxel and doxorubicin-containing adjuvant chemotherapy: The experience of the BIG 02-98 trial. *Breast Cancer Res Treat* 2010; 119:145–153.

50. Majed B, Moreau T, Senouci K et al. Is obesity an independent prognosis factor in woman breast cancer? *Breast Cancer Res Treat* 2008;111:329–342.

51. Maiti B, Kundranda MN, Spiro TP et al. The association of metabolic syndrome with triple-negative breast cancer. *Breast Cancer Res Treat* 2010;121:479–483.

52. Suba Z. Triple-negative breast cancer risk in women is defined by the defect of estrogen signaling: Preventive and therapeutic implications. *Onco Targets Ther* 2014;7:147–164.

53. Liu B, Fan Z, Edgerton SM et al. Metformin induces unique biological and molecular responses in triple negative breast cancer cells. *Cell Cycle* 2009; 8:2031–2040.

54. Davis AA, Kaklamani VG. Metabolic syndrome and triple-negative breast cancer: A new paradigm. *Int J Breast Cancer* 2012;2012:809291.



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